

Tracking wetland community evolution using Diptera taxonomic, functional and phylogenetic structure

AMELIE GREGOIRE TAILLEFER*  and TERRY A.

WHEELER[†]  Department of Natural Resource Sciences, McGill University, Ste-Anne-de-Bellevue, Quebec, Canada

Abstract. 1. Different processes drive spatial variation in community composition. Standard measures of composition are useful in species-based conservation and ecology, but they may be less informative in the context of evolutionary history and functional diversity. Functional and phylogenetic approaches are increasingly used to test mechanisms driving biodiversity patterns.

2. We studied 28 families of flies (Diptera) with a range of functional characteristics in three wetland classes (bogs, swamps, marshes) in Quebec, Canada. We examined taxonomic, phylogenetic and functional structure of communities and assessed whether rarity is deterministic or stochastic. Beta- and phylobeta-diversity were also examined for relatedness to local environmental conditions, patch area, and/or surrounding landscape.

3. Phylogenetic community structure analyses had high value and complementarity to standard measures. Environmental filtering acted on bog communities during assembly, as they emerged from a slow peat accumulation process and the plant composition is characteristic as few species can survive in these acidic and low nutrient conditions. Subsequently, community assembly happened randomly. Neutral processes of community assembly are more important in marshes and swamps, as dispersal limitation explained species abundance dynamics of small and common Diptera species. The assembly of marsh communities is a balance between neutral processes and environmental filtering, while assembly in swamps can be seen as neutral.

4. Clustering increased with environmental extremes, indicating environmental filtering. Rare species tended to be less closely related to common species. They have unique habitat requirements, and the high diversity is maintained by temporal turnover of species with similar traits filtered by the environment.

Key words. Clustering, community assembly, environmental factors, environmental filtering, neutral processes, rare species, stochasticity.

Introduction

What are the processes underlying spatial variation in species richness and community composition? Standard measures of species richness and composition (e.g. Gotelli & Colwell, 2001; Magurran, 2004) focus mainly on alpha- and beta-diversity. Although these metrics are important

in conservation and ecology, they can be uninformative in the context of evolutionary history and functional diversity, as they treat all species as ecologically equivalent and evolutionary independent (Swenson *et al.*, 2012). A promising complementary modern approach has been built on the incorporation of phylogenetic data in ecological analyses (Webb, 2000; Webb *et al.*, 2002; Cavender-Bares *et al.*, 2009).

More than twenty years ago, Palmer (1994) compiled a list of more than 120 hypotheses predicting drivers of species diversity, and undoubtedly, this list would be even longer today. What drives turnover of species across space

*Correspondence: Amélie Grégoire Taillefer, Department of Natural Resource Sciences, McGill University, Ste-Anne-de-Bellevue, Quebec, Canada. E-mail: amelie.gregoire-taillefer@mail.mcgill.ca

[†]Deceased.

and time is one of the oldest questions in ecology, and no general consensus has arisen to explain the patterns observed. Several alternative theories have been proposed. The niche theory dates back to Grinnell (1924) who defined an 'ecological or environmental niche' as the distributional unit where a species can live. A few years later, Elton (1927) developed a concept of niche in terms of food habits of a species. Later, Hutchinson (1957) defined the notion of ecological niche as a multidimensional space of suitable environmental variables for the survival of a species. More recently, Webb (2000) and Cavender-Bares *et al.* (2006) incorporated the phylogenetic structure of a community into ecological deterministic processes in the niche model that influence the community pattern. The concept of neutral theory developed by Bell (2000) and Hubbell (2001) assumes an equivalence between individuals (i.e. differences have no functional significance) and stochastic processes in species coexistence such as birth, death, speciation and dispersal. The continuum hypothesis (Gravel *et al.*, 2006) proposed to unify the niche and neutral models, where the importance of deterministic and stochastic processes varies along a continuum. The model focuses on local community scale, where recruitment is a lottery between species competitive ability (niche processes) and dispersal limitations (neutral processes). Another dynamic model (Jabot *et al.*, 2008) proposed a mathematical framework in which to measure the recruitment limitation defined as the influence of dispersal limitation and post-dispersal filtering of species. Vellend (2010) organised the large number of ecological theories into four distinct processes to provide a general conceptual framework in community ecology: selection, drift, speciation and dispersal.

Species assemblages are affected by different mechanisms from the mosaic of the habitat types to the composition of the land cover surrounding their habitats. This matrix of surrounding heterogeneous habitats may drive colonisation-extinction dynamics by limiting dispersal (e.g. Ricketts, 2001; Revilla *et al.*, 2004), by modifying resource availability (Grundel & Pavlovic, 2007) and by increasing edge effects (Saunders *et al.*, 1991). The impact of the landscape matrix may be different depending on anthropogenic disturbance, amount and configuration of natural patches (Kennedy *et al.*, 2010). Moreover, communities occur on gradients of environmental variables and these interactions (e.g. facilitation, predation, and competition) along with evolutionary relationships, and species functional traits may determine community structure (Davis *et al.*, 1998; McGill *et al.*, 2006).

As conservation of biodiversity and biotic communities is becoming a high priority in threatened and vulnerable habitats such as wetlands, incorporating species traits (functional structure), phylogenetic relationships, landscape structure, and spatial scales may help us understand the ecological and evolutionary factors that determine community assembly. Few studies have investigated patterns of diversity that consider a few of these approaches simultaneously (e.g. traits and phylogeny: Cavender-Bares

et al., 2006; Vamosi & Vamosi, 2007; phylogeny and environment: Horner-Devine & Bohannan, 2006; and traits, scale, and environment: Arnan *et al.*, 2012).

Wetlands are one of Canada's most characteristic and recognisable habitats. Canada's wetlands comprise an estimated 24% of the world's wetlands and about 14% of Canada's surface. They are important ecologically as habitats, nutrient and carbon sinks, water supplies, etc. The global loss of wetlands on Canada's landmass is estimated to be 20 million hectares since 1800, and degradation has affected millions of others. Wetlands have been lost, fragmented, and drained for horticultural peat mining, forestry, urbanisation, and agriculture (Lavoie & Rochefort, 1996; Waddington & Price, 2000). Most of the loss has taken place near urban areas, where access is easy. Bogs are peat-covered wetlands, low in nutrients and acidic with pH between 3 and 5. They have characteristic vegetation dominated by *Sphagnum* mosses (Gorham & Janssens, 1992; Payette & Rochefort, 2001). Marshes are rich in nutrients as they are periodically inundated. They contain shallow water, 15 to 90 cm in depth with pH > 5, which allows emergent herbaceous vegetation to grow. Swamps are dominated by trees and shrubs where the water at or below the surface is rich in nutrients and usually neutral with pH around 7.2 (National Wetlands Working Group, 1997).

We studied Schizophora (Diptera) taxa in three classes of wetlands (bogs, marshes, swamps) in the Montreal metropolitan area and surroundings, Quebec, Canada. Diptera are separated into two major sub-orders: Lower Diptera and Brachycera. Within Brachycera, the clade Schizophora contains more than 80 families, and 60 families in the Nearctic region of the 150 families of Diptera that are recognised (McAlpine *et al.*, 1981; McAlpine, 1989; Yeates & Wiegmann, 2005). Schizophora is monophyletic and contains the majority of the family-level diversity among Diptera. Moreover, Schizophora represents a recent rapid radiation of lineages (Woodley *et al.*, 2009; Lambkin *et al.*, 2013); it covers a broad range of life-history and biological traits, is abundant and species rich in wetlands (Keiper *et al.*, 2002; Beaulieu & Wheeler, 2005; Grégoire Taillefer & Wheeler, 2010; Omelková *et al.*, 2013) and is overlooked in most ecological studies. This study characterised the diversity, distribution, ecological roles, and local rarity of Diptera present in wetlands isolated in an anthropogenic matrix and assessed the mechanisms that structure community phylogenetics and beta-diversity.

We examined whether (i) wetland Schizophora communities exhibit taxonomic, phenotypic, and phylogenetic structure and (ii) whether rare species within wetland habitats are ecologically dissimilar to common species or whether commonness and rarity are essentially stochastic. Also, we examined (iii) how taxonomic, phylogenetic, and functional compositions are related to local environmental conditions, patch area, and/or surrounding landscape. We hypothesised that species composition is related to local conditions and diversity follows a gradient from poorer

(ombrotrophic) to richer (minerotrophic) habitats, specifically bogs to marshes to swamps. Phylogenetic clustering will be more important than overdispersion, as only certain species possess the necessary adaptations to survive in these harsh habitats. The most stressful habitat, bogs, acts as a stronger ecological filter than the less-stressful swamp habitat. Beta-diversity is related to dispersal barriers such as distance between sites and urban development.

Materials and methods

Four wetlands in each class were sampled for *Schizophora* in summer 2013 (Fig. 1). Wetlands were classified following the Canadian Wetland Classification System (National Wetland Working Group, 1997). Wetlands were chosen from high-resolution digital aerial photographs available from Ducks Unlimited Canada for the territory of the Montreal metropolitan area (Beaulieu *et al.*, 2010) and from personal communication with experts. To standardise the wetland type, open raised bogs with pH < 4.5 dominated by *Sphagnum* mosses, ericaceous shrubs and a tree cover of <30% were chosen. Marshes were covered with permanent shallow water with pH between 5.5 and 7.4. Swamps were covered entirely by deciduous trees, and the water table was at the ground surface at the beginning of the season. Sampling was conducted weekly for six weeks from 8 July 2013 to 11 August 2013 with pan traps and sweeping on an area of 30 m × 30 m in the centre of each wetland if accessible. Sweep samples were collected on three transects of 20 m in each site. The location of sweep transects within each site differed from week to week. Three transects of four pan traps were placed 10 m apart on a 30-m transect in the centre of each site and emptied every 7–8 days. Traps consisted of yellow plastic bowls (4.5 cm deep and 12.5 cm diameter)

placed in the soil with their upper rim flush with the ground surface and filled with a 50% solution of propylene glycol and water, with a drop of liquid detergent as a wetting agent.

Insects were preserved in 95% ethanol for subsequent DNA extraction. Small flies were dried using hexamethyldisilazane; larger flies were transferred into ethyl acetate, and then pinned and air-dried. Specimens were deposited in the Lyman Entomological Museum (McGill University, Ste-Anne-de-Bellevue, QC). *Schizophora* taxa were identified to named species or to numbered morphospecies.

The definition of rarity used in this study is based on three indirect rarity measures: (i) restriction to a wetland habitat (restricted: found in only one wetland type; intermediate: found in two wetland types; broad: found in three wetland types); (ii) frequency of occurrence (rare: only found in one site; infrequent: found in two to five sites; intermediate: found in six to nine sites; widespread: found in 10 to 12 sites); and (iii) number of specimens collected (rare: <0.6% of total catch; common: ≥0.6% of catch). To separate rare and common species, an inflection point criterion was used. The rank abundance curve for the total abundance was examined, and the inflection point was where the curvature changed (Siqueira *et al.*, 2012). Each species was also assigned to a category in seven biological traits: larval feeding habit, specialisation to wetlands, size, generalist/specialist, voltinism, overwintering forms, and habitat preference for oviposition (Table S1).

Water pH and conductivity measurements were taken with Hanna pocket EC/TDS and pH Tester at two locations in the 30 × 30 m area in each site in week one in the same weather conditions and time of day. Fluctuation of water-table depth over the season was estimated visually with the PVC tape discoloration method (Belyea, 1999). The following ground cover attributes were surveyed in five 1 m × 1 m quadrats placed at each corner

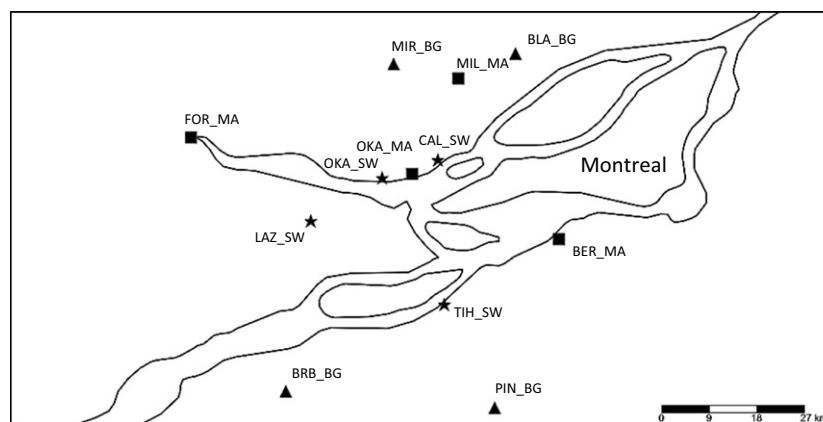


Fig. 1. Map of the 12 study locations in the Montreal metropolitan area and surroundings, Quebec, Canada. Stars: swamps; squares: marshes; triangles: bogs. (CAL_SW: Pointe-Calumet swamp; TIH_SW: St-Thimothé swamp; OKA_SW: Oka National Park swamp; LAZ_SW: St-Lazare swamp; BER_MA: Ile St-Bernard marsh; OKA_MA: Oka National Park marsh; MIL_MA: Mirabel Parc du Domaine Vert marsh; FOR_MA: Pointe-Fortune marsh; BLA_BG: Blainville bog; PIN_BG: Pin-Rigide bog; BRB_BG: Ste-Barbe Large Tea Field bog; MIR_BG: Mirabel bog). Map created with SimpleMapp (www.simplemapp.net).

and in the middle of the 30 m × 30 m area: per cent plant species cover, litter cover, canopy cover, and open water cover. Landscape composition and wetland area were measured using the Earth Observation for Sustainable Development of Forests land cover data sets of circa year 2000 at a spatial resolution of 25 m (Wulder *et al.*, 2008) and QuantumGIS version 1.8.0 (QGIS, 2012). Open water, exposed land, urban development, low vegetation, wetland, forest and agriculture surrounding each wetland were measured within a circle with a 2 km radius because Diptera have been found to respond to the surrounding matrix at this spatial scale (Meats & Smallridge, 2007; Savage *et al.*, 2011) (Table S2).

Statistical analyses

Diptera diversity and community structure. Diptera communities in each site were characterised by relative abundance, observed species richness and diversity from samples of pan traps and sweeping pooled by week and throughout the season. Individual-based accumulation curves at the global scale (all sites) and individual-based rarefaction curves at the local scale (individual sites) were computed to assess whether saturation was obtained with the methods of collection smoothed by 1000 repetitions of randomised sampling using the vegan package (Oksanen *et al.*, 2015) in R version 3.2.2 (R Core Team, 2015). Significance of differences in overall relative abundance, observed species richness, rarefied species richness and expected species richness (Chao) among wetland type were determined based on ANOVA *F* tests.

The habitat association of each species was examined using indicator species analysis performed with the function *multipatt* in the package *indicspecies* (De Cáceres & Legendre, 2009). The association of each species with individual or combinations of classes in each wetland type was applied. The significance of species association was assessed with a permutation test using 999 permutations. Only species with ≥10 individuals and an indicator value (IndVal) ≥85% were considered.

A non-metric multidimensional scaling (NMDS) ordination was used to assess differences among sites and weeks in relation to their Diptera community composition using the function *metaMDS* of the vegan package. A Hellinger transformation (Legendre & Gallagher, 2001) was used on the species data to give low weight to rare species. A permutational multivariate analysis of variance based on a Bray–Curtis distance matrix was used to assess significance of differences among habitats using the function *Adonis* of the vegan package. Ground cover variables were reduced with principal component analysis (PCA) using the correlation matrix (Dormann *et al.*, 2013), and principal components (PC) were used in subsequent analyses. Significant PC were identified using the Kaiser–Guttman criterion (Yeomans & Golder, 1982). Variables with loadings larger than 0.32 were forming the clusters of interest (Booth *et al.*, 1994) (Table 1). Latitude and

longitude were transformed into the corresponding coordinates in *X* (east–west) and *Y* (north–south) distances using the function *geoXY* in the package *SoDA* (Chambers, 2008). To reduce the number of environmental variables in the subsequent analyses and to avoid autocorrelation between them, first all remaining environmental variables were observed with scatterplots and Pearson's correlation indices were calculated for all combination of variables (Dormann *et al.*, 2013). The resultant matrix included the PC1 to 4, *X* and *Y* coordinates, % tree cover, % water above surface, conductivity, plant richness and all landscape variables. Environmental variables were standardised prior to subsequent analyses. The function *envfit* of the vegan package was used to examine the significant relationship between the

Table 1. Ground cover variables reduced with principal component analysis. Variables with loadings larger than 0.32 in principal components (PC) 1 to 4 are in bold.

Variables	PC1	PC2	PC3	PC4
<i>Abies balsamea</i>	−0.31	0.43	0.00	0.47
Bare soil	−0.31	0.08	−0.01	−0.53
<i>Betula alleghaniensis</i>	−0.31	0.43	0.00	0.47
<i>Betula populifolia</i>	0.23	0.00	−0.01	−0.02
<i>?Calla palustris</i>	−0.31	0.43	0.00	0.47
<i>Carex oligosperma</i>	0.46	0.16	−0.40	0.02
<i>Carex viridula</i>	−0.31	0.43	0.00	0.47
<i>Cephalanthus occidentalis</i>	−0.23	−0.12	−0.01	−0.18
<i>Chamaedaphne calyculata</i>	0.52	0.17	−0.28	0.03
<i>Cornus stolonifera</i>	−0.17	−0.09	−0.02	0.11
Dead wood	−0.37	−0.06	−0.02	0.02
<i>Drosera rotundifolia</i>	0.28	0.07	−0.05	0.03
<i>Equisetum pratense</i>	−0.46	0.49	0.00	0.03
<i>Eriophorum vaginatum</i>	0.49	0.09	0.46	0.06
<i>Fraxinus nigra</i>	−0.23	0.29	0.02	−0.55
<i>Hydrocharis morsus-ranae</i>	−0.28	−0.61	−0.12	0.07
<i>Kalmia angustifolia</i>	0.37	0.05	0.55	0.05
<i>Larix laricina</i>	0.14	−0.03	0.09	−0.01
<i>Ledum groenlandicum</i>	0.29	0.05	0.40	0.05
Litter	−0.47	−0.03	0.11	−0.08
<i>Lysimachia terrestris</i>	0.34	0.15	−0.54	−0.01
<i>Onoclea sensibilis</i>	−0.49	0.35	0.01	−0.28
Open pool	−0.51	−0.33	−0.11	0.33
<i>Osmunda regalis</i>	−0.28	0.25	0.02	−0.57
<i>Phalaris arundinacea</i>	−0.10	−0.40	−0.08	0.12
<i>Phragmites australis</i>	−0.17	−0.09	−0.02	0.11
<i>Pinus rigida</i>	0.34	0.15	−0.54	−0.01
<i>Pleurocarpus moss</i>	−0.65	0.15	−0.05	0.04
<i>Polytrichum</i> spp.	0.58	0.13	0.34	0.05
<i>Sagittaria latifolia</i>	−0.18	−0.40	−0.08	0.02
<i>Salix pedicellaris</i>	0.20	0.01	0.33	0.01
<i>Salix pyrifolia</i>	0.34	0.15	−0.54	−0.01
<i>Sphagnum</i> spp.	0.66	0.19	−0.05	0.04
<i>Spiraea alba</i>	0.34	0.15	−0.54	−0.01
<i>Spirodela polyrhiza</i>	−0.14	−0.50	−0.10	0.10
<i>Toxicodendron radicans</i>	−0.37	0.49	0.02	−0.23
<i>Typha angustifolia</i>	−0.14	−0.52	−0.10	0.11
<i>Vaccinium ?angustifolium</i>	0.29	0.05	0.40	0.05
<i>Vaccinium myrtilloides</i>	0.20	0.01	0.33	0.01

environmental variables and species occurrence, using 999 permutations.

Phylogeny reconstruction. Due to a lack of resolved phylogeny for Brachycera (Diptera), a dated tree of Schizophora was generated with palaeontological calibration dates. A single gene tree was used as a proxy for a species tree (Joly *et al.*, 2014). Sequences of the 658-bp fragment of the cytochrome *c* oxidase subunit I (COI-5P) gene of Eremoneura, Cyclorrhapha and Schizophora taxa for this phylogeny were obtained from the Barcode of Life Database (BOLD) (Ratnasingham & Hebert, 2007) or from our own sequenced specimens. Three hundred and fifty-six specimens of 139 species not available in BOLD were sent for DNA barcoding at the Canadian Centre for DNA Barcoding, Guelph, Ontario, using *C_LepFolF/C_LepFolR* primer sets (Hernández-Triana *et al.*, 2014). Sequences of *Dolichopus brevipennis* (Dolichopodidae), *Sphaerophoria philanthus* (Syrphidae), *Chelipoda truncata* (Empididae), *Hypocera ehrmanni* (Phoridae), and *Lonchoptera furcata* (Lonchopteridae) were used to root the tree (outgroup). Monophyly was constrained for superfamilies, families, sub-families and genera in which there were recent and well-supported, multi-gene or morphological phylogenies for Schizophora: selected Acalyptratae superfamilies (Yeates & Wiegmann, 2005); Schizophora and Calyptratae (Lambkin *et al.*, 2013); Sarcophagidae (Kutty *et al.*, 2010; Pape *et al.*, 2011); *Ravinia* (Sarcophagidae) (Giroux *et al.*, 2010; Piwczynski *et al.*, 2014); *Boettcheria* (Sarcophagidae) (Piwczynski *et al.*, 2014); Luciliinae and Polleniinae (Calliphoridae) (Kutty *et al.*, 2010); Phytomyzinae (Agromyzidae) (Scheffer *et al.*, 2007); Scathophagidae (Bernasconi *et al.*, 2000); Milichiidae (Brake, 2000); Sciomyzidae, Sciomyzini, Tetanocerini (Tóthová *et al.*, 2013); Tephritidae (Han & Ro, 2009); Drosophilidae (Yassin, 2013); and Chloropinae (Chloropidae) (Brake, 2000).

A Bayesian approach was used for phylogenetic analyses because it allows for both phylogenetic relationship assessments and divergence time estimates for speciation rates (Espíndola *et al.*, 2012). Trees were generated with the program BEAST v1.7.5 (Drummond *et al.*, 2012), and the output was examined via Tracer v1.6 (Rambaut *et al.*, 2014). The MUSCLE (Edgar, 2004) multiple alignment software was used to produce an aligned data matrix. Because no suitable calibration points or mutation rates for the study taxa exist, we used JModelTest2 (Guindon & Gascuel, 2003; Darriba *et al.*, 2012) and Akaike information criterion to determine which model of nucleotide substitution best-fit the alignment. This gene-based quantitative analysis was chosen with a fossil-based analysis because the fossil record is very limited.

In all analyses, strong priors were set on the age of two nodes corresponding to Schizophora (LogNormal: mean = 3, SD = 0.78, offset = 70) and Chloropidae (LogNormal: mean = 3, SD = 0.7, offset = 42) according to palaeontological and molecular data (Nardi *et al.*, 2010; Wiegmann *et al.*, 2011). The appropriate model

selected by JModelTest2 was GTR + I + G, and no partitioning by codon was used. A branching prior was set under a Yule process model, and a relaxed molecular clock was assumed using a LogNormal distribution of rates (Drummond *et al.*, 2006). The analysis was performed twice for a MCMC chain length of 40 million generations (10 000 echo states, 4000 log parameters, 10% burn-in) (Fig. 2).

Phylogenetic and functional community structure. Phylogenetic signal for categorical traits was measured using the fitDiscrete function in the package Geiger (Harmon *et al.*, 2008). Pagel's lambda was used as a tree transformation that assesses the degree of phylogenetic signal within the trait. Maximum-likelihood estimates of lambda were compared with a model with no phylogenetic signal where the tree topology was transformed to one giant basal polytomy that has all internal branch lengths multiplied by 0 (i.e. $\lambda = 0$) (Freckleton *et al.*, 2002). Significance of the models was assessed by comparing them with a chi-square distribution. For the continuous trait (size), the *K* statistic (Blomberg *et al.*, 2003) in the package picante (Kembel *et al.*, 2010), that compares the observed signal in a trait to the signal under a Brownian motion model of trait evolution on a phylogeny, was measured to assess whether the pattern was conserved ($K > 1$), convergent ($K < 1$) or random ($K = 0$). To test for independence in the phylogenetic signals, a generalised least squares (GLS) method was then used to test for relationships between significant traits, subsequently taking phylogenetic relatedness into account with phylogenetic generalised least squares (pGLS) method.

To determine whether Diptera communities and functional groups are more or less phylogenetically related than expected by chance several indices were calculated. Faith's phylogenetic diversity metric (PD) (Faith, 1992) was calculated for each community. Then, a phylogenetic distance matrix from the tree generated was produced with the *cophenetic* function from the package ape (Paradis *et al.*, 2004). The mean pairwise distance (MPD) was calculated as the mean phylogenetic distance among all pairwise combinations of co-occurring species, and the mean nearest neighbour distance (MNTD) was calculated as the mean phylogenetic distance to most closely related species for all co-occurring species (Webb, 2000; Webb *et al.*, 2008) with the package picante. To analyse the whole community structure, these two metrics were measured among individuals (weighted by abundance) and among species to determine the effect of rare species. For functional groups, only abundance-weighted measures were calculated, because individuals, not species, are filtered and compete in a community (Cornwell & Ackerly, 2009; Fine & Kembel, 2011). Null models were used to determine whether the phylogenetic and functional compositions of communities differ significantly from that expected by chance (Gotelli, 2000). A null model where taxon labels were randomised 999 times at the tip of the tree, while maintaining species abundance and tree

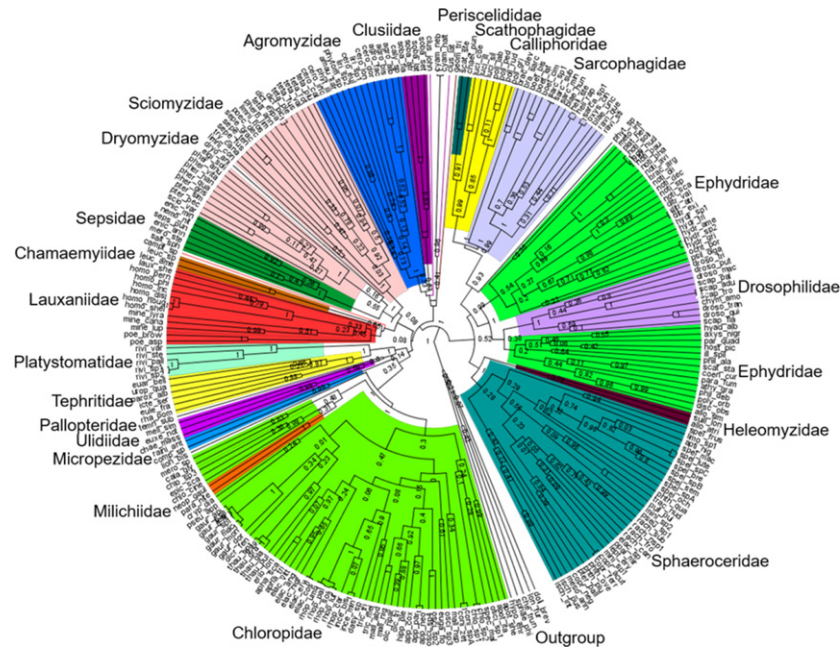


Fig. 2. Hypothesised phylogenetic relationships, based on CO1, among Schizophora (Diptera) species collected in the 12 wetlands. Colours represent families, branch lengths represent divergence time estimates, and numbers on nodes represent posterior probabilities. Species codes can be found in Table S1. [Colour figure can be viewed at wileyonlinelibrary.com].

topology was used (Kembel & Hubbell, 2006). These 999 null measurements were used to calculate measures of standardised effect size for MPD and MNTD: net relatedness index (NRI) and nearest taxon index (NTI) multiplied by -1 (observed value minus the mean value of the 999 null values divided by the standard deviation of the 999 null values), respectively, and probabilities. Negative NRI/NTI values and high quantiles ($P > 0.95$) indicate phylogenetic evenness, while positive values and low quantiles ($P < 0.05$) indicate phylogenetic clustering, relative to the null model. To determine whether the mean NRI and NTI values overall and per wetland class are significantly different among habitats and from zero, t -tests were performed. Trait diversity within communities was calculated in the same way as phylogenetic relatedness, with standardised effect size as a measure of trait dissimilarity among co-occurring species.

Phylobeta-diversity. To evaluate the phylogenetic and functional beta-dispersion of Diptera communities within and between the three wetland classes, beta-phylogenetic differences weighted by abundance were tested using *comdist* (the average MPD for each species in a sample to all species in another sample) and *comdistnt* (the average MNTD for all species in a sample to the nearest neighbours in another sample) using the package *picante*. Bray–Curtis dissimilarity among all sites was calculated as an abundance-weighted measure of beta-diversity, and pairwise dissimilarities between traits using the function *daisy* with the Gower's distance measure in the package *cluster* (Maechler *et al.*, 2015) were calculated as a

measure of trait dispersion among all sites. The output matrices were used in NMDS ordinations with the *monoMDS* function to determine the differences in phylogenetic community structure using the package *picante*. The function *envfit* was also used to examine the significant relationship between the environmental variables and phylogenetic dispersion, using 999 permutations. To determine whether beta-diversity and phylobeta-diversity were positively spatially autocorrelated, the Mantel statistic in the *ade4* package (Dray & Dufour, 2007) was calculated with spatial distance. To assess correlations between species co-occurrence and phylogenetic distances, the Mantel statistic was also performed between a Bray–Curtis dissimilarity matrix and beta-phylogenetic and functional dissimilarity matrices.

Results

Alpha-taxonomic structure

A total of 6602 specimens representing 295 species belonging to 28 families were used in analyses (Table S3). Muscidae and Anthomyiidae were not included in this study, as keys for species identification are mostly unavailable for many genera in these families. Diptera assemblages varied from 196 to 1645 individuals in each sampling site (Table 2). Accumulation curves for individual sites showed that species diversity was far from saturation at each site based on collecting effort (Fig. 3), although Chao indices suggest that our collecting effort

retrieved between ~50 and 90% of Schizophora diversity (Table 2). In contrast, when considering the global set of sites, the individual-based accumulation curve reached an asymptote at 11 sites. The number of species increased rapidly with the addition of the first 10 sites, but once ~11 sites were considered, the addition of more sites resulted in a small addition of new species. Global extrapolated species richness with Chao was evaluated to 413.8 ± 29.9 individuals, meaning between 65 and 76% of species were collected. *Scatella stagnalis* Fallén (Ephydriidae) was the most abundant species accounting for 12% of total abundance and was the most widely distributed species, being found in all sites. The next most abundant was *Leptocera erythrocerca* (Becker) (Sphaeroceridae) (8%), followed by *Rachispoda limosa* (Fallén) (Sphaeroceridae), *Incertella bispina* (Malloch) (Chloropidae) and *Elachiptera nigriceps* (Loew) (Chloropidae), together accounting for 4% of the total specimens. Indicator species analysis revealed that 13 species were significantly associated with one type of wetland and that two were associated with two types (Table 3).

Beta-taxonomic structure

When testing for the influence of habitat type with ANOVAS, relative abundance ($F = 1.63$, $P = 0.25$), observed species richness ($F = 2.14$, $P = 0.173$), rarefied species richness ($F = 0.69$, $P = 0.52$) and extrapolated species richness (Chao) ($F = 1.375$, $P = 0.3$) (Table 2) were not significantly different between wetland types. High turnover of species was observed from week to week in marshes and bogs as supported by NMDS; although in swamps, species compositions from week to week were overlapping (Fig. 4a). For the functional groups, the abundance and richness of each trophic group were not

Table 2. Abundance (N), species richness (Sobs) rarefied species richness $\pm$ standard error (Sest) and extrapolated species richness (Chao) $\pm$ standard error (Percentage contribution of observed species richness to expected species richness) (Sext) for each of the 12 wetland sites. Rarefied species richness was standardized at 196 individuals (PIN_BG). See Figure 1 legend for site codes.

Site	N	Sobs	Sest	Sext
CAL_SW	532	73	49.2 ± 3.1	114.6 ± 20.6 (63%)
TIH_SW	1646	105	38.4 ± 3.7	143.0 ± 16.6 (73%)
OKA_SW	903	86	43.6 ± 3.5	126.6 ± 18.0 (68%)
LAZ_SW	234	62	57.6 ± 1.8	88.0 ± 13.1 (70%)
BER_MA	258	52	46.5 ± 2.0	78.3 ± 15.5 (67%)
OKA_MA	545	56	33.6 ± 2.9	88.5 ± 17.2 (63%)
MIL_MA	1037	91	46.2 ± 3.5	124.2 ± 16.1 (73%)
FOR_MA	252	33	30.5 ± 1.4	36.7 ± 4.5 (90%)
BLA_BG	237	30	27.5 ± 1.4	60.3 ± 22.7 (50%)
PIN_BG	196	40	40.0 ± 0.0	57.1 ± 12.0 (70%)
MIR_BG	290	67	56.2 ± 2.6	89.4 ± 12.2 (75%)
BRB_BG	468	72	45.5 ± 3.3	141.0 ± 33.4 (51%)

significantly different between wetland habitat, and the overall distribution was similar in each habitat. Saprophages were the most abundant and species rich, followed by phytophages, predators and parasites (except in bogs where parasites were more diverse than predators). Rare species were less abundant, but more species rich compared to common species and the abundance and richness decreased from small- to large-sized species (Fig. 5).

Community composition shown by NMDS ordination (stress = 0.081) indicated distinct assemblages between the three types of wetlands (Adonis: $R^2 = 0.4$, $P = 0.001$). Species assemblages in each wetland class were associated mainly with local conditions: tree cover ($R^2 = 0.94$, $P = 0.001$), PC1 ($R^2 = 0.9$, $P = 0.001$), conductivity ($R^2 = 0.85$, $P = 0.002$), water above surface ($R^2 = 0.77$, $P = 0.002$), and/or PC2 ($R^2 = 0.74$, $P = 0.004$) (*Hydrocharis morsus-ranae*, *Typha angustifolia*, *Spirodela polyrhiza*, *Equisetum pratense*, etc. (Table 1)). Only open water ($R^2 = 0.68$, $P = 0.01$) and exposed land ($R^2 = 0.49$, $P = 0.04$) surrounding the habitats had a lower positive association with marsh and bog, respectively (Fig. 4b).

Alpha-phylogenetic structure

The test of phylogenetic signal using Pagel's lambda showed that the restriction to a wetland type ($\lambda = 0.61$, $\ln L = -238.11$, $\ln L_{\lambda=0} = -240.898$, $P = 0.018$) was significantly non-randomly distributed over the tree. Blomberg's K value for size ($K = 0.55$, $P = 0.001$) showed that closely related species resemble each other less in size than expected under the Brownian motion model of trait evolution. The GLS of size as a function of wetland type (ANOVA $F = 0.83$, $P = 0.44$) and the pGLS ($F = 0.7$, $P = 0.5$) models were both non-significant.

As expected, Faith's PD showed the same pattern as species richness as it is highly correlated with species richness, where none of the pairwise comparisons of wetland classes for the entire pool of species or by functional group displayed significant differences ($P > 0.05$). The phylogenetic structure of Diptera communities was never evenly distributed, only a few sites showed clustering and a large proportion showed randomness when compared to phylogenetically random communities generated by the null model. For all sites pooled together, mean NRI value for individuals (2.24, $P = 0.016$) and taxa (2.68, $P = 0.0012$) and mean NTI value for individuals (1.4, $P = 0.01$) and taxa (1.99, $P = 0.0001$) were significantly clustered. At the level of the entire tree, the bog habitat was significantly clustered for co-occurring species ($\text{mean NRI} = 3.9$, $P = 0.026$) but not for co-occurring individuals ($\text{mean NRI} = 2.07$, $P = 0.1$). The other two wetland habitats were not significantly different from a random distribution ($\text{mean NRI} = 0$, $P > 0.05$). Phylogenetic clustering towards the tip of the tree was significant for marsh habitat for individuals ($\text{mean NTI} = 2.29$, $P = 0.04$) and species ($\text{mean NTI} = 2.57$, $P = 0.01$) (Fig. 6). The swamp habitat contains species randomly distributed at each level

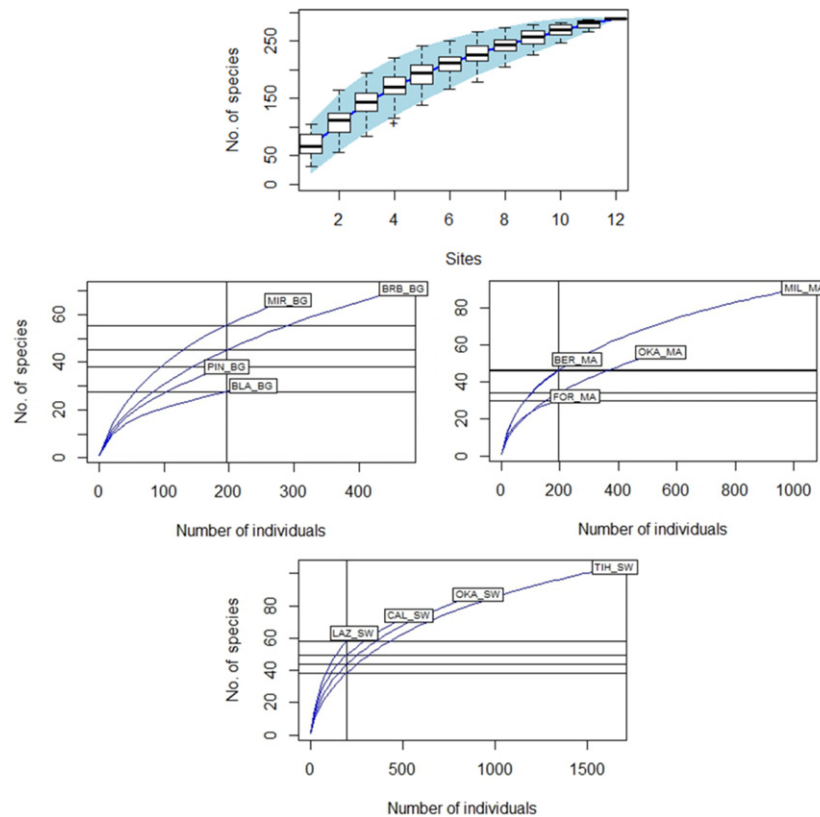


Fig. 3. Individual-based accumulation curves at the global scale (all sites) (above) and individual-based rarefaction curves at the local scale (individual sites) (below) were computed to assess whether saturation was obtained with the methods of collection smoothed by 1000 repetitions of randomised sampling. Species counts at 196 individuals for rarefied estimates are indicated by the vertical black line. [Colour figure can be viewed at wileyonlinelibrary.com].

Table 3. Indicator species analysis of Brachycera species for bogs, marshes and swamps. Only species with ≥ 10 individuals and with an indicator value (IndVal) greater than 85 and a significant ($P \leq 0.05$) indicator value are shown.

Species	Number of individuals	Significant Association	IndVal	P-value
<i>Homoneura sheldoni</i> (Coquillett) (Lauxaniidae)	65	Bog	92	0.018
<i>Euaresta bella</i> (Loew) (Tephritidae)	36	Bog	89	0.018
<i>Malloewia abdominalis</i> (Becker) (Chloropidae)	122	Bog	90	0.050
<i>Malloewia nigripalpis</i> (Malloch) (Chloropidae)	17	Bog	90	0.050
<i>Notiphila avia</i> Loew (Ephydriidae)	59	Marsh	89	0.018
<i>Notiphila olivacea</i> Cresson (Ephydriidae)	82	Marsh	87	0.022
<i>Spelobia</i> sp.A (Sphaeroceridae)	20	Swamp	95	0.020
<i>Spelobia luteilabris</i> (Rondani) (Sphaeroceridae)	10	Swamp	100	0.009
<i>Pterogramma palliceps</i> (Johnson) (Sphaeroceridae)	47	Swamp	96	0.009
<i>Drosophila transversa</i> Fallén (Drosophilidae)	37	Swamp	88	0.023
<i>Chymomyza amoena</i> (Loew) (Ephydriidae)	66	Swamp	96	0.009
<i>Athyroglossa granulosa</i> (Cresson) (Ephydriidae)	82	Swamp	93	0.025
<i>Ditrichophara exigua</i> Cresson (Ephydriidae)	147	Swamp	90	0.045
<i>Rhopalopteron carbonarium</i> (Loew) (Chloropidae)	140	Bog/Marsh	95	0.035
<i>Elachiptera nigriceps</i> (Loew) (Chloropidae)	244	Marsh/Swamp	96	0.007

in the phylogeny and for every functional group. The mean NRI for the functional groups showed that bog communities were more closely related than expected for

large species (2.5, $P = 0.016$), parasites (2.62, $P = 0.01$) and rare species (2.7, $P = 0.049$) overall and by week (2.06, $P = 0.0009$). The mean NRI value for rare species

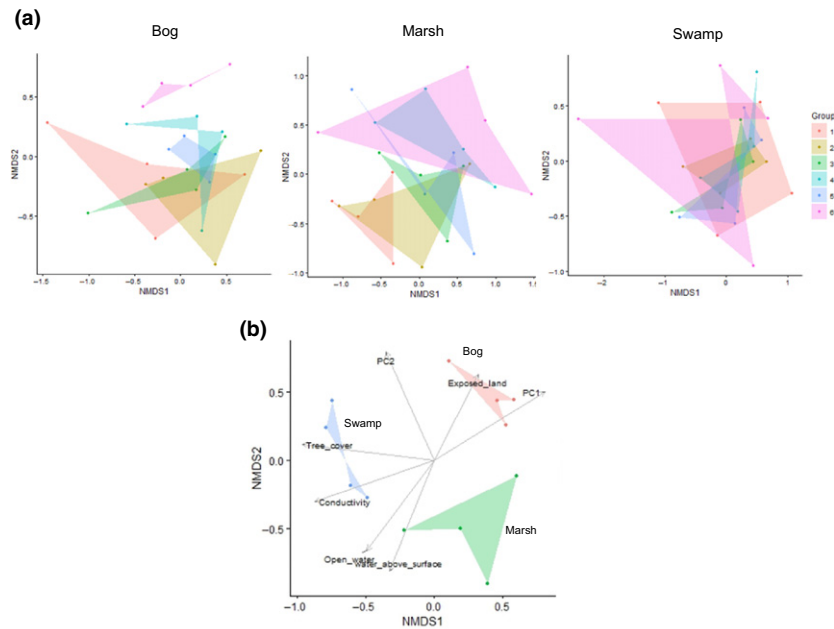


Fig. 4. Non-metric multidimensional scaling ordination of Diptera species collected (a) by week in each wetland type and (b) overall in the three types of wetlands. Arrows represent vectors of the seven variables that explain Diptera species composition. [Colour figure can be viewed at wileyonlinelibrary.com].

in marshes by week was also significantly clustered (1.15, $P = 0.0009$). The mean NTI values in marsh habitat showed clustering for medium species (1.76, $P = 0.026$), saprophages (1.72, $P = 0.01$) and rare species (1.97, $P = 0.018$) overall and by week (0.96, $P = 0.00023$) and in bogs for large species (1.46, $P = 0.034$), rare species by week (1.44, $P = 0.00027$) and restricted species (1.99, $P = 0.04$). Standardised effect sizes for all traits pooled were not significantly different from a random expectation.

Beta-phylogenetic structure

Mantel correlation between beta- and phylobeta-diversity was only significantly correlated with spatial distance for small ($r = 0.32$, $P = 0.032$) and common ($r = 0.31$, $P = 0.01$) species beta-NTI values. All Mantel correlations for taxonomic Bray–Curtis versus phylogenetic diversity and trait dissimilarities were significant. Adonis analyses showed significant differences between types of wetland for all NMDS ordinations except for beta-diversity in traits MPD ($R^2 = 0.24$, $P = 0.1$) (Fig. 7) and beta-diversity in common species MNTD/MPD ($P > 0.05$). Overall MPD and MNTD phylogenetic beta-diversity were influenced each by a subset of environmental variables influencing community structure. MPD beta-diversity was affected in decreasing order by conductivity ($R^2 = 0.83$, $P = 0.001$), PC1 ($R^2 = 0.77$, $P = 0.02$), surrounding open water ($R^2 = 0.67$, $P = 0.016$) and tree cover ($R^2 = 0.61$, $P = 0.01$). MNTD beta-diversity was

influenced in decreasing order by PC1 ($R^2 = 0.9$, $P = 0.001$), tree cover ($R^2 = 0.82$, $P = 0.003$), PC2 ($R^2 = 0.75$, $P = 0.005$), conductivity ($R^2 = 0.68$, $P = 0.011$), surrounding exposed land ($R^2 = 0.62$, $P = 0.025$), surrounding open water ($R^2 = 0.59$, $P = 0.025$) and water above surface ($R^2 = 0.58$, $P = 0.013$). Trait beta-diversity was influenced by a subset of environmental variables influencing beta-MPD/MNTD (Fig. 7).

Discussion

Taxonomic structure

This study is one of the few to use a highly diverse insect group in the context of community phylogenetics and the first study of community phylogenetics in a diverse community of Diptera. Our study, although small in scale and with unequal sampled number of individuals between sites, had a species accumulation curve that leveled-off at 11 sites. This indicates that a significant portion of diversity was captured within the time frame and with the two complementary sampling methods. Schizophora species are usually well represented in wetlands, but overlooked in most studies of terrestrial Diptera diversity (Castella *et al.*, 1994; Barták & Roháček, 1999; Pollet, 2001). There is probably a large number of wetland associates still to discover, and the status of a large number of species is unknown because of insufficient ecological knowledge. Savage *et al.* (2011) collected

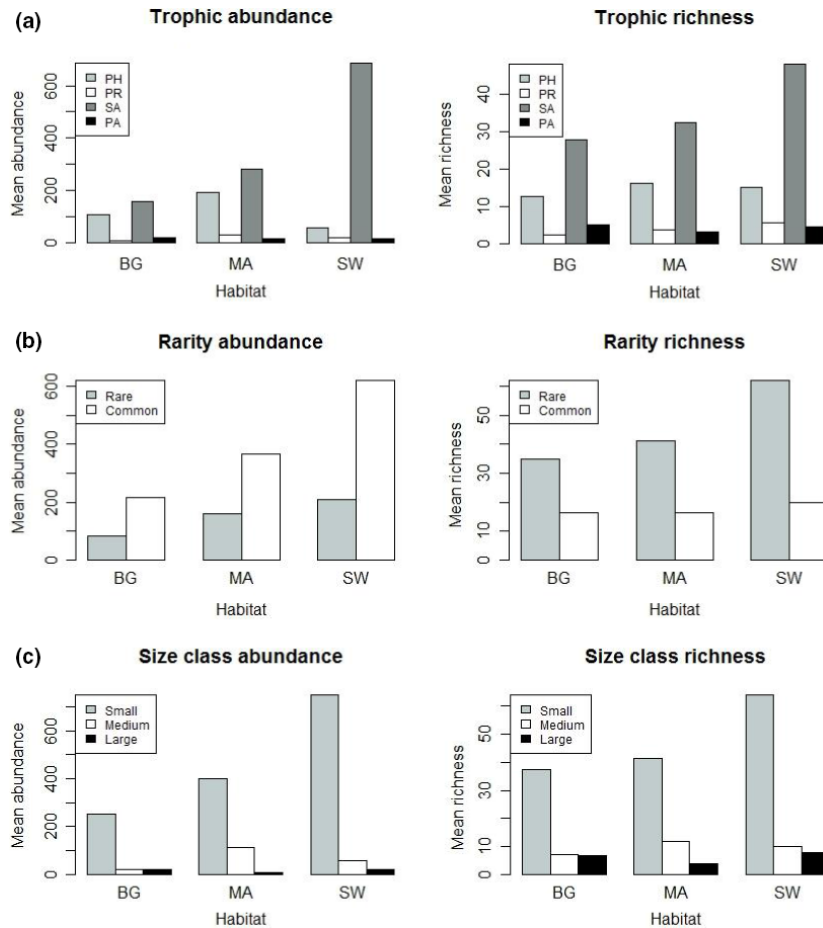


Fig. 5. Changes in (a) Trophic mean abundance and richness; (b) rarity mean abundance and richness; and (c) size class mean abundances and richness of Diptera between bogs (BG), marshes (MA) and swamps (SW) for phytophages (PH), predators (PR), saprophages (SA) and parasites (PA).

6700 specimens and 381 species of Schizophora in six temperate bogs in southern Quebec and Vermont. Blades and Marshall (1994) surveyed the terrestrial arthropod fauna associated with substrate (pan trap collection) of Southern Ontario peatlands and found 50 families of Diptera, representing 522 species. The most species rich families of higher flies were in decreasing order: Sphaeroceridae, Dolichopodidae, Tachinidae, Sarcophagidae, Empididae, Syrphidae, Chloropidae and Ephydriidae. Using multiple families of Diptera representing different trophic and ecological groups, community responses are more likely to represent all possible responses to changes in environmental variables. The results are concordant with evidence, indicating that the different types of wetlands support a similar species richness, but differences in community composition are driven mainly by local conditions and on a lower level by surrounding open water and exposed lands. There was greater similarity within habitat type than among habitats, even the indicator species were found in different families among the

three wetland types. All indicator species (Table 1) except one (*Euaresta bella*) were saprophagous. In the bog habitat, only one species is considered a characteristic inhabitant: *Homoneura sheldoni* (Lauxaniidae) (Miller, 1977). *Euaresta bella* (Tephritidae) is a widespread endemic species to North America feeding on the common ragweed *Ambrosia artemisiifolia* L. The two *Malloewia* (Sphaeroceridae) species are saprophagous in a variety of habitats and widespread in eastern North America. They seem to prefer open habitats such as wetlands and grasslands (T.A. Wheeler, pers. comm.). The two *Notiphila* (Ephydriidae) species associated with marshes are detritus feeders in anaerobic conditions and are obligatory associates with roots of wetland plants (Larson & Foote, 1997). In swamps, it seems that all indicator species with a known ecology are facultative in wetlands. The only known breeding record of *Athyroglossa granulosa* (Ephydriidae) is from skunk cabbage (*Symplocarpus foetidus*) that grows in wetlands in North America (Grimaldi & Jaenike, 1983).

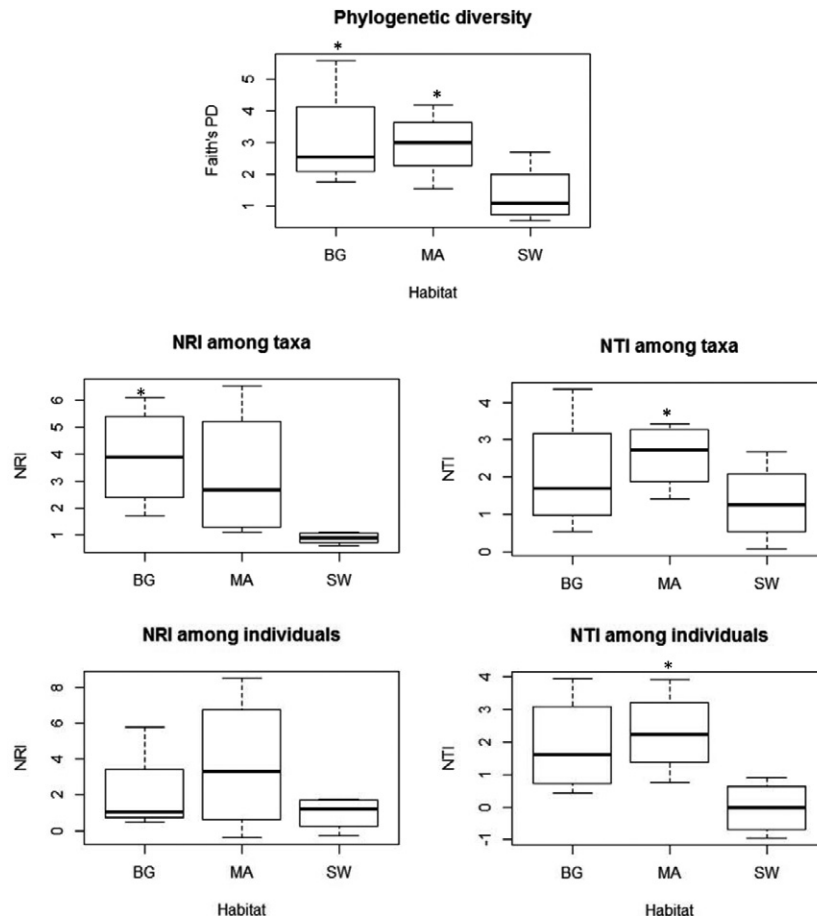


Fig. 6. Distribution of mean $\pm$ standard deviation of phylogenetic diversity, net relatedness index (NRI) and nearest taxon index (NTI) of Diptera across the three wetland types (BG: bog, MA: marsh, SW: swamp). *Significant deviation from 0 (t-test) at $p < 0.05$.

Phylogenetic structure

Environmental filtering acts on the regional scale, as species were phylogenetically segregated in the different wetlands. This is reflected in the phylogenetic signal of the wetland type, the distinct phylobeta-diversity and in the different species composition present in each wetland type. At the site scale, phylogenetic clustering tree-wide is argued to result from either radiation or different colonisation patterns of habitats in the distant past (Hardy & Senterre, 2007; Webb *et al.*, 2008). Ecological-scale community assembly from a species pool that survived the last glaciation is a much more plausible explanation. Most species that survived the last glaciation, the Wisconsin glacial maximum between 22 000 and 14 000 BP, retreated into new niches or found refugia south of the ice sheets (Scudder, 1979; Beatty & Provan, 2010). By 10 000 BP, the Laurentide ice sheet covering Quebec was reduced by 35%, and by 7000 BP, the ice sheet had almost completely receded. The bogs emerged from a very slow peat accumulation process, since the early to middle Holocene (9000–6000 BP) in southern Canada (Ovenden, 1990; Vitt, 1994), which

coincided with the Hypsithermal, the warmest time interval of the current interglacial (Pielou, 1991). These warmer and drier conditions permitted dispersal of flora and fauna northwards (Hewitt, 2000). Consequently, the current distribution of species in bog communities, particularly those with low dispersal range, is still influenced by this past event. During community assembly, environmental filtering first acted on species where plant assemblages (PC1) influenced which traits allowed species establishment. The plant composition is characteristic with only a few species that can survive in these acid and low nutrient conditions. Therefore, the characteristic bog plant species are locally abundant, but do not occur in the surrounding environments. Environmental filtering mostly affected large, parasitic and rare fly species. Several Diptera species are considered peatland specialists (Teskey & Burger, 1976; Miller, 1977; Farkas & Brust, 1986; Marshall, 1994), likely because specialist species survived glaciation in suitable habitats in the southern refugium, and then dispersed northwards as the ice receded. In accordance with our hypotheses, we found indications that the bogs support closely related species filtered by harsher environmental conditions, as the

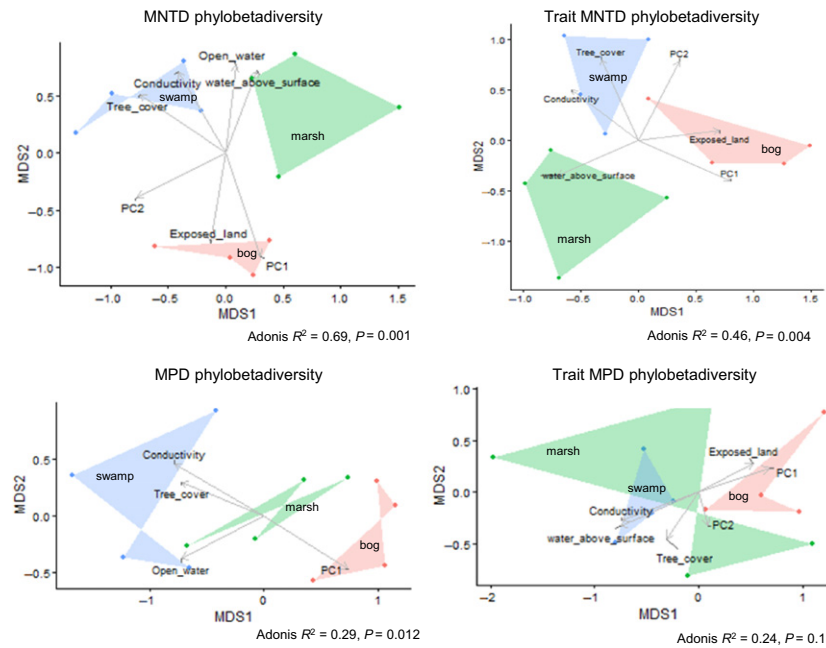


Fig. 7. Non-metric multidimensional scaling ordination of species and trait phylogenetic beta-diversity of mean pairwise distance (MPD) and mean nearest taxon distance (MNTD) of Diptera in the three types of wetlands. Arrows represent vectors of the variables that explain phylogenetic turnover in community composition. [Colour figure can be viewed at wileyonlinelibrary.com].

entire food web is associated with the water chemistry; peat is acidic in nature and has low available minerals such as nitrogen, calcium, magnesium and phosphorus (Coulsen & Butterfield, 1978). This finding is consistent with Silver *et al.* (2012), who found similar results for aquatic Diptera in temporary wetlands. As they start to dry, environmental stresses increase and so increasing phylogenetic clustering is observed. Several authors have suggested that environmental factors at the landscape level may explain more variation in arthropod species composition than local factors (e.g. Økland *et al.*, 1996; Schweiger *et al.*, 2005). In this study, however, the uniqueness of the bog habitat at the landscape level may explain why local variables are more important than surrounding land use for Diptera community composition. Recent turnover of species probably comes from the surrounding exposed lands (e.g. non-vegetated, non-developed, sediments, beaches, burnt area) where Diptera species have environmental requirements suitable for bog habitats. This is in contrast to Savage *et al.* (2011) who found that species compositions in temperate bogs were mostly affected by the presence of a forest buffer in between urban development and agricultural land.

A random structure could result if opposing patterns of clustering and overdispersion are found within the same community (Hardy & Senterre, 2007; Mayfield & Levine, 2010). At the site scale and at every taxonomic scale, however, overdispersion was not found. Neutral processes are more important in marshes and swamps, as dispersal limitation may explain species abundance dynamics of small and common species within habitat type as they are affected by the inter-site distances (Condit *et al.*, 2002). The assembly

of marsh communities may be seen as recent processes with a balance between neutral processes and environmental filtering, where species emigrate from surrounding lentic habitats and are filtered by the variation in the water level. Species must be adapted to survive the variation in the hydrological cycle or disperse to more suitable habitats (Murkin & Batt, 1987). The assembly of swamp habitats can be seen as solely neutral processes as this habitat is more heterogeneous and shares more characteristics with the surrounding environment such as forests, for the tree cover, and water bodies, for the higher mineral content (National Wetlands Working Group, 1997).

The detailed life-history and ecological traits that confer habitat preference for Diptera species in bogs and marshes were, however, not measured, as the functional status of a large number of species is unknown because of insufficient ecological knowledge. Nonetheless, the degree of phylogenetic relatedness can possibly be used as a proxy for ecological similarity as all significant traits showed clustering in NRI and NTI values. Poff (1997) analysed quantitatively evolutionary lability of traits and found that life history (e.g. voltinism, synchronisation of emergence) was more evolutionary constrained than ecological traits (e.g. trophic habit, thermal preference), probably because life-history traits are associated with the reproductive success, thus the persistence of species. Behavioural (e.g. specialisation) and feeding-related traits have been found to be involved in habitat selection at smaller scales, while size, dispersal ability and the number of generations per year have been associated with larger scales (Larsen & Ormerod, 2010). No phylogenetic signal for diet overall

was found, although one could potentially be found at a lower level as these broad categories can be divided into several subcategories (e.g. saprophages can be divided into detritus feeders, coprophages, bacterial feeders, secondary invaders of damaged plants). Even if a phylogenetic signal is not detected in a community as a whole, it might exist within a clade (Losos, 2008).

Taxonomic versus phylogenetic structure

The three types of wetlands studied are distinct in terms of their taxonomic and phylogenetic community structure and trait composition within particular terminal clades (although the clades are spread out on the entire phylogeny, resulting in similar trait MPD phylobeta-diversity across wetland types). The relative abundance, species diversity and functional trait distribution are, however, similar among habitats. Trends of the changes in beta-diversity with the wetland gradient are similar when either measured taxonomically or phylogenetically. Without examining phylogenetic relatedness, we would have concluded that high species turnover and random functional turnover indicate stochasticity in species turnover and community assembly, owing to dispersal limitation (Swenson *et al.*, 2011). Phylogenetic analyses in complement with ecological metrics have been shown to report an effect of treatments when classical metrics (abundance, diversity) do not (Kelly *et al.*, 2015). No change in alpha-diversity, abundance or functional diversity along this environmental gradient could be explained by functionally similar species replacing each other in the communities and that flies perceived their environment at a smaller scale than humans.

Rarity effect

Rarity is an important concept in community ecology, and no consensus has been reached about the mechanisms underlying abundance of species. According to several authors, rare species are assembled by niche differentiation and specialisation, and/or temporal resource partitioning which differ from environmental preferences or dispersal limitation of common species (e.g. Kunin & Gaston, 1993; Chesson, 2000; Magurran & Henderson, 2003). The niche differentiation hypothesis suggests that three predictions must be followed. First, a phylogenetic signal of abundance is expected. Second, rare species should contribute more to community phylogenetic diversity, and third, phylogenetic dissimilarity should be observed within and between assemblages of common and rare species (Mi *et al.*, 2012). Our findings support this hypothesis. On average, species present in one wetland type tended to be distantly related to species present in two or three types as a phylogenetic signal was found. Rare species assemblages had higher cumulative phylogenetic diversity for nine of twelve sites than common species assemblages which were all randomly distributed.

Rare species assemblages showed clustering in marsh and bog habitats and by week, while common species distribution was always random and influenced by dispersal limitation. In accordance with Thompson and Townsend (2006), this shows that local ecological factors and dispersal limitation play equally important roles in macroinvertebrate distribution in temperate wetlands. The rarity effect revealed hidden patterns that different processes affect the occurrence of species and the local dominance as suggested by Chalmardier *et al.* (2015). The different patterns of spatial partitioning along this wetland gradient suggest that niche differentiation by temporal partitioning of the active season may contribute to the high diversity of rare species in these communities.

Size is usually used as a proxy for dispersal capacity; here, we can consider the use of a morphological surrogate for ecology as small species are affected by spatial distances. Low phylogenetic signal in a trait is usually considered as evolutionary lability (Blomberg *et al.*, 2003; Silvertown *et al.*, 2006), although the evolutionary process underlying this lability is difficult to detect (Revell *et al.*, 2008). Errors in phylogenetic topology as well as estimation of species means will downwardly bias the calculation of phylogenetic signal (Blomberg *et al.*, 2003; Ives *et al.*, 2007). In this study, the hypothesised phylogeny was constructed with one gene, although most of the species were clustering by family. The average size for species was measured from two mounted individuals, which could have brought some variation and lowered the phylogenetic signal.

Conclusions and conservation implications

Integrating taxonomic and functional responses with phylogenetic relatedness provided additional and complementary information about mechanisms driving biodiversity patterns in different temperate wetland habitat types. Although our data set contains only four sampling sites in each wetland type, the results reported here confirm that Schizophora communities exhibit phylogenetic structure in wetlands and phylogenetic clustering meaning that environmental filtering increases with environmental extremes. We found evidence that rare wetland Schizophora species are mainly affected by local ecological factors and that the high diversity is maintained by temporal turnover of species with similar traits filtered by the environment. Species composition differed highly among the three wetland types, suggesting that descriptions of local habitats and community composition, and the response of local species to habitat changes should be considered when making interventions. Also, maintaining a diversity of pristine wetland habitats within a region is of major importance for species diversity. Even the Oka marsh and swamp that were separated by 5 km, without barriers to dispersal, supported each a unique assemblage. Also, results suggest that human interventions aimed at conserving and restoring wetland structure and function need to vary the focus of attention in the different wetland types. When neutral

assembly processes are important, restoration projects may require intervention such as propagule addition or removal to consider stochasticity (Grégoire Taillefer & Wheeler, 2013). On the other hand, when niche processes are more important, restoration projects should focus on desired habitat properties for species of interest. The wide range of wetland site size (6.7–161.6 ha) did not influence the diversity and abundance of Diptera, which was similar to results found by Savage *et al.* (2011). Area has been found to influence predacious long-legged fly communities in reed marshes (Decler *et al.*, 2015), so the effect may vary with size and specialisation and be obscured in these functionally diverse communities. Local characteristics are the major determinants of composition. We could argue that even small fragments of wetlands are of major importance to maintain high diversity and abundance of Diptera. This is in line with Perović *et al.* (2015) who showed that maintaining a higher level of landscape complexity (within 250–1000 m of the habitat patch) with many small important habitat patches for butterflies was maintaining increased taxonomic and functional diversity. More configurational heterogeneous landscapes are shown to support more microhabitats thus more species with different ecological functions (Devictor & Jiguet, 2007; Fahrig *et al.*, 2011). Given the rapid disappearance of wetlands and their invaluable roles for carbon sequestration, habitat for endangered species, water quality and supply, this study bears further examination based on additional data, including at a larger scale and with other types of organisms.

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Supporting Information

Additional Supporting Information may be found in the online version of this article under the DOI reference: doi: 10.1111/icad.12271:

Table S1. Traits assigned to each Schizophora species and morphospecies collected from each of the 12 wetland sites.

Table S2. Coordinates, surrounding landscape, local and chemical variables and ground cover attributes at each of the 12 wetland sites.

Table S3. Abundance of Schizophora species and morphospecies collected from each of the 12 wetland sites. Species in grey were excluded from analyses.

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